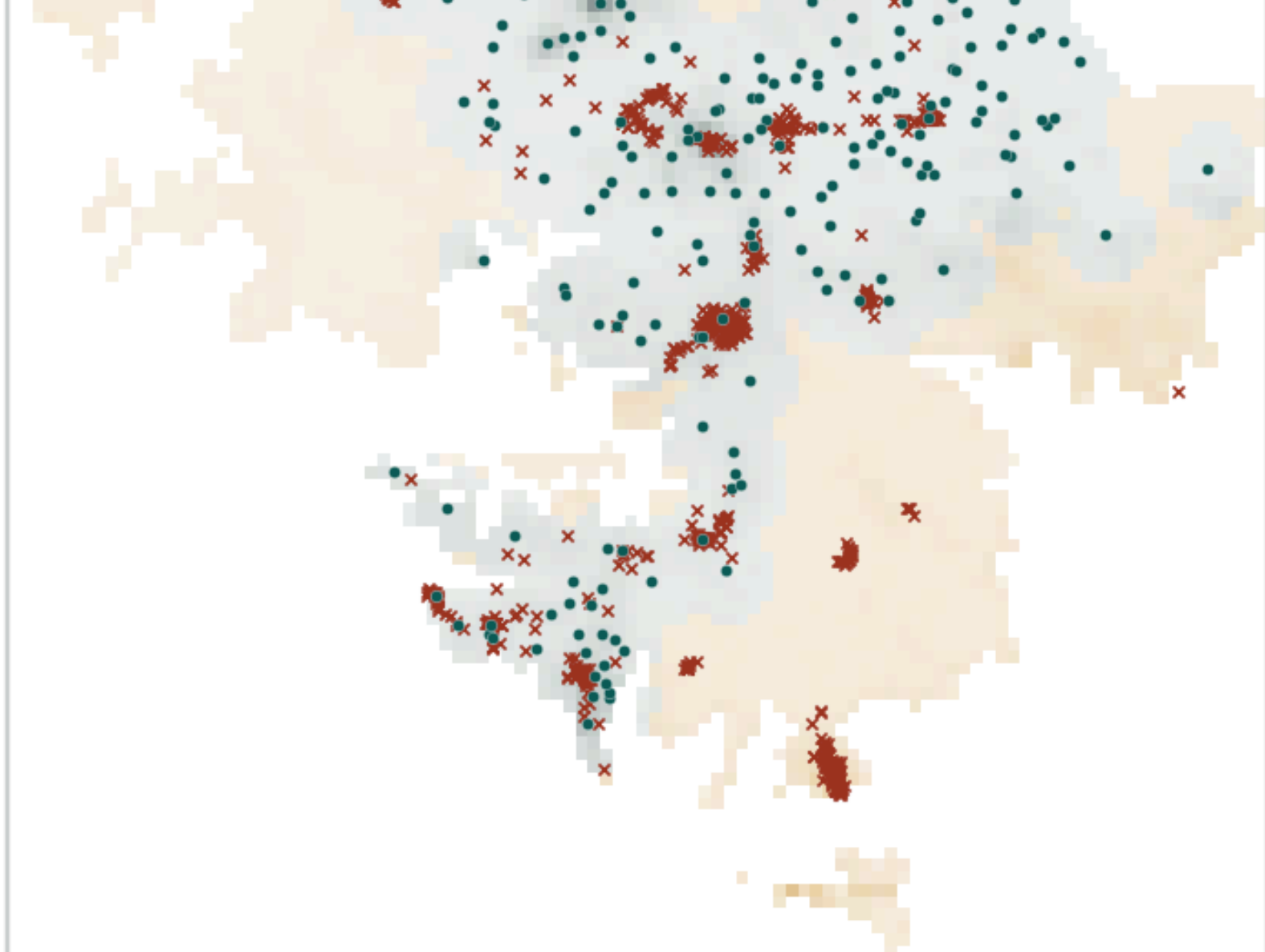




Rural Water Point Siting Assessment: Mayuge District

Uganda · September 2026

OPEN SITING AGENT

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Rural Water Point Siting Assessment: Mayuge District

Document Control

Assessment	Rural Water Point Siting Assessment: Mayuge District
Jurisdiction	Mayuge District, Uganda (UGA)
Subject	Placement of 14 additional water points under a fixed budget
Issued	2026-09-04T15:59:05Z
Run identifier	mayuge-water-20260904T155905Z
Provenance hash	ac15a4eae9de9c52
Prepared by	An autonomous analysis agent operating on open data
Reviewed by	Jadey (see Chapter 5)
Supersession	This assessment supersedes no prior issue. Each run is issued under its own identifier and no run is revised in place.
Maintenance	Regenerated by replaying the manifest recorded in Appendix C. Any change to a source query, a cleaning rule or a retrieval date changes the provenance hash.

Status of this Assessment

This assessment is advisory. It was produced from published open data without field verification. Land availability, tenure, water table, community consent and construction feasibility are outside the data used here and remain matters for the reviewing officer. Recommendations in Chapter 4 should be read together with the limitations at Section 1.4 and the data anomalies at Appendix A.

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Chapter 1: Introduction

Chapter 1 states the purpose of this assessment, how it should be read, what it covers, and the limits within which its findings hold. It closes with the data sources on which every figure in the assessment rests.

Purpose

This assessment identifies where 14 additional water points should be placed in Mayuge District so as to bring the largest number of people currently without service inside a water point service radius, and states what each of those placements would achieve.

How to Read this Assessment

Chapter 2 describes the current state of service in the district. Chapter 3 states how the assessment was carried out, including every retrieval made and every cleaning rule applied. Chapter 4 gives the recommendation. Chapter 5 records the review, including each point at which a reviewing officer overruled the analysis and what that decision cost. Chapter 6 tests how far the recommendation depends on assumptions that can reasonably be changed.

Appendix A records properties of the source data that would have changed a figure had they been read differently. Appendix B is the retrieval record. Appendix C is the reproduction record.

Scope and Applicability

This assessment covers Mayuge District only. Coverage is computed against water point records held in the source register; infrastructure that is not recorded appears here as absence of service. The method is not specific to this district or this domain, and the same procedure applies without modification to health facility access, disease control and environmental monitoring.

Limitations

1. **The register is not a census.** Coverage is computed against recorded water points only. Unmapped infrastructure appears as absence of service.
2. **Records age.** The median record for this district is 1.9 years old. Findings describe the surveyed state of the network, not necessarily its state today.
3. **Distance is straight-line.** Terrain, watercourses and footpath networks are not modelled, so real walking distances exceed the radius used here.
4. **Costs are not modelled.** Rehabilitating an existing point and constructing a new one draw equally on the budget. Chapter 6 reports what changes when only rehabilitation is permitted.
5. **No field verification was performed.** No site in Chapter 4 has been visited.

Decisions This Assessment Rests On

Certain judgements cannot be derived from data: what distance counts as served, whose need a fixed programme answers, and whether a register of a given age is fit to direct capital spending.

Each is recorded below with the name of whoever took it and the reason they gave. The analysis refuses to run without them rather than assuming one.

Table 1. Decisions Recorded for This Assessment

Decision	Value	Recorded by	Reason given
service_radius_m	1000	Jadey	1000 m Mayuge already reports against,
coverage_basis	walking_time	Jadey	walking time : Mayuge walk around water rather than across it
objective	max_coverage	Jadey	198,832 unserved is too many to spend a small programme on the margins
budget	10	Jadey	we have varied budget, we want to find an ideal number
data_currency_accepted	yes	Jadey	conditional, some will have been repaired and some of the 354 will have failed
coverage_tolerance	0.05	Jadey	198,832 unserved is too many to spend a small programme on the margins
review_floor	6.5	Jadey	don't be too strict
equity_accepted	yes	Jadey	198,832 unserved is too many to spend a small programme on the margins

Every judgement above was recorded by a person before the analysis ran.

Data Sources

Table 2. Sources Used in this Assessment

Source	Endpoint	Licence	Retrieved
WPdx+	https://data.waterpointdata.org/resource/eqje-vguj.json	CC BY 4.0	2026-09-04
GADM	https://geodata.ucdavis.edu/gadm/gadm4.1/json/gadm41_UGA_1.json.zip	GADM licence, free for academic and non-commercial use	2026-09-03
WorldPop	https://hub.worldpop.org/rest/data/pop/wpgp	CC BY 4.0	2026-09-03
MAP friction surface	https://data.malariaatlas.org/geoserver/Accessibility/ows	CC BY 4.0	2026-09-04

Chapter 2: Situation Overview

Chapter 2 describes service coverage in Mayuge District as recorded at the time of retrieval, and states the size of the population that a capital programme in this district would be seeking to reach.

Recorded Service Coverage

An estimated 570,781 people live within the area of interest. 371,949 of them, 65.2%, live within about 21 minutes of walking over the modelled terrain of a water point recorded as serving. The remaining 198,832 people, 34.8% of the district, do not.

The register holds 1,251 water point records for this district after cleaning. 354 are recorded as serving and 897 are not, which is 72% of the recorded stock.

Table 3. Recorded Coverage Before Intervention

Category	People	Share of district
Within the service radius of a serving water point	371,949	65.2%
Beyond the service radius	198,832	34.8%
Total resident population	570,781	100.0%

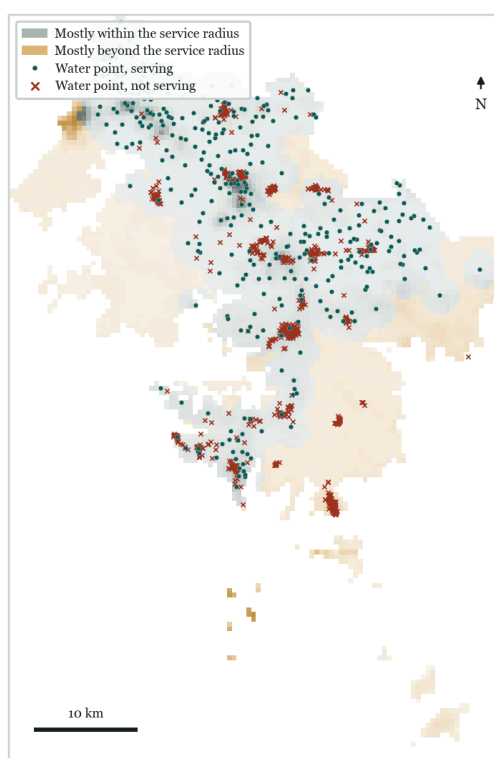


Figure 1. Recorded Water Points and Population Beyond the Service Radius

Sources: WPdx+; GADM; WorldPop; MAP friction surface. Coordinates WGS 84. Warm shading indicates population beyond the service radius of any water point recorded as serving.

The Gap Expressed in People

A percentage conceals the size of the problem. 198,832 people is the figure that bears on planning: it is the population a capital programme in this district would be attempting to reach.

Chapter 3: Assessment Methodology

Chapter 3 states how the assessment was carried out. It covers retrieval and cleaning, the construction of the demand surface and the candidate set, the selection method and the guarantee it carries, and the validation applied before findings were released.

Retrieval and Cleaning

Each source is governed by a handbook file that declares its endpoint, the meaning of its fields, and the cleaning rules to be applied in order. Each rule reports its own drop count, so no record leaves the pipeline unaccounted for.

Table 4. Records Retrieved and Retained

Source	Returned	Retained	Discarded
WPdx+	1,378	1,251	127
GADM	58	58	0
WorldPop	15,912	8,992	6,920
MAP friction surface	5,670	5,670	0

Records were discarded from WPdx+ for the following reasons: 127 flagged duplicate by WPdx.

Why Discarded Records Are Itemised

A register that returns a record does not necessarily hold a distinct, current observation. Where a source ships duplicate rows and superseded observations in the same response, coverage computed on the raw response counts settlements twice and treats retired observations as current. Every rule that removes a record is declared in advance and reports how many it removed, so the difference between the returned and retained counts above is fully explained.

Coordinate Reference System

All distance, radius and separation computations are carried out in EPSG:32636 (WGS 84 / UTM zone 36N). Area of interest spans 0.43 degrees of longitude and is centred at 0.279, 33.497, which falls in UTM zone 36N. UTM is conformal and preserves local distance to about one part in 2,500 within a zone, which is well inside the tolerance of a service-radius analysis.

Over 400 sampled pairs of demand cells, projected distance departs from great-circle distance by 0.317% at the median and 0.595% at most, which on a 1000 m service radius is about 6 m.

Why Distance Is Not Computed in Degrees

Computing distance directly from latitude and longitude is among the most common errors in applied spatial analysis, because a degree of longitude is not a fixed ground distance and shrinks towards the poles. A service radius is precisely the kind of claim that must hold in metres on the ground, so the analysis projects first and measures second. The figures above state how far the chosen projection departs from the sphere here, rather than assuming the difference away.

Demand Surface

Population is taken from a gridded estimate at approximately 100 m resolution, read over the area of interest and aggregated into blocks coarser than the source and finer than the service radius, so that aggregation does not bias coverage. 5,153 demand cells result, together holding 570,781 people.

Candidate Set

2,095 candidate locations were evaluated. Two kinds were admitted: an existing water point recorded as not serving, which could be rehabilitated, and a regular lattice of new locations at 750 m spacing. Lattice points already inside an existing service area were removed before selection. Recommendations are constrained to lie at least 1000 m apart.

Selection Method

Candidates are scored by the population they would newly bring inside the service radius and selected greedily.

On a monotone submodular coverage function greedy selection is within a factor of $1 - 1/e$, about 63 per cent, of the optimum.

Coverage Is a Union, Not a Sum

A settlement within reach of two water points is served once, not twice. Coverage throughout this assessment is therefore the union over selected locations. Summing the population served by each location separately overstates coverage, and the independent recount reported at Section 5.2 states by how much it would have done so here.

Validation

Before findings were released, the plan was checked by a separate process that can reject it but cannot amend it, and which does not read the reasons recorded in Chapter 5. A single rejection stops the assessment being issued. The results of those checks are at Section 5.2.

Chapter 4: Findings

Chapter 4 gives the recommended placements, the population each would reach, and the basis on which each was selected.

Recommended Placements

Table 5. Recommended Water Point Placements, in Selection Order

No.	Site	Type	Location (WGS 84)	Newly covered	Nearest serving
1	S-001	greenfield	0.5090, 33.3442	12,365	2.1 km
2	S-002	greenfield	0.3258, 33.6677	7,240	4.3 km
3	S-003	greenfield	0.3393, 33.6340	6,715	3.5 km
4	S-004	rehabilitate	0.1889, 33.5601	6,100	7.5 km
5	S-005	greenfield	0.2715, 33.5800	5,076	5.4 km
6	S-006	greenfield	0.2308, 33.6003	4,775	8.4 km
7	S-007	greenfield	0.2172, 33.5194	4,609	3.0 km
8	S-008	greenfield	0.3597, 33.6879	4,589	2.8 km
9	S-009	greenfield	0.2308, 33.5666	4,533	4.8 km
10	S-010	greenfield	0.2647, 33.6070	4,207	8.5 km
11	S-011	greenfield	0.2986, 33.5666	4,167	3.8 km
12	S-012	greenfield	0.4819, 33.3374	4,093	4.6 km
13	S-013	greenfield	-0.0135, 33.6609	4,051	30.7 km
14	S-014	greenfield	0.1426, 33.5733	3,946	11.8 km

Each row's coverage is additional to every row above it. The programme must therefore be built from the top of the table down; taking rows out of order does not produce the stated coverage.

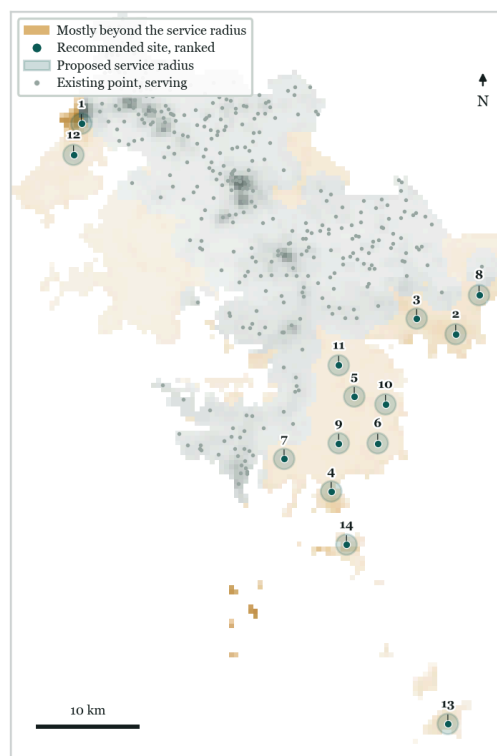


Figure 2. Recommended Placements and Their Service Radii

Coverage Achieved

Selecting the 14 placements above brings a further 76,465 people inside the service radius, raising recorded coverage from 65.2% to 78.6%.

Table 6. Marginal and Cumulative Coverage by Number of Placements

Placements built	Newly covered	Cumulative covered	Share of district
1	12,365	384,314	67.3%
2	7,240	391,554	68.6%
3	6,715	398,268	69.8%
4	6,100	404,368	70.8%
5	5,076	409,444	71.7%
6	4,775	414,219	72.6%
7	4,609	418,828	73.4%
8	4,589	423,417	74.2%
9	4,533	427,950	75%
10	4,207	432,157	75.7%
11	4,167	436,324	76.4%
12	4,093	440,417	77.2%
13	4,051	444,468	77.9%
14	3,946	448,414	78.6%

Determining How Many to Build

Returns fall away as placements are added. The first placement reaches 12,365 people; the 14th reaches 3,946. A budget decision taken against Table 5 rather than against a round number will reach more people for the same expenditure.

Basis for Each Placement

S-001	New site covering 12,365 people currently beyond the service radius; nearest working point 2.1 km away.
S-002	New site covering 7,240 people currently beyond the service radius; nearest working point 4.3 km away.
S-003	New site covering 6,715 people currently beyond the service radius; nearest working point 3.5 km away.
S-004	Rehabilitate the existing point covering 6,100 people currently beyond the service radius; recorded as non-functional; technology on record: Public Tapstand; nearest working point 7.5 km away.
S-005	New site covering 5,076 people currently beyond the service radius; nearest working point 5.4 km away.
S-006	New site covering 4,775 people currently beyond the service radius; nearest working point 8.4 km away.
S-007	New site covering 4,609 people currently beyond the service radius; nearest working point 3.0 km away.
S-008	New site covering 4,589 people currently beyond the service radius; nearest working point 2.8 km away.
S-009	New site covering 4,533 people currently beyond the service radius; nearest working point 4.8 km away.
S-010	New site covering 4,207 people currently beyond the service radius; nearest working point 8.5 km away.
S-011	New site covering 4,167 people currently beyond the service radius; nearest working point 3.8 km away.
S-012	New site covering 4,093 people currently beyond the service radius; nearest working point 4.5 km away.
S-013	New site covering 4,051 people currently beyond the service radius; nearest working point 30.7 km away.
S-014	New site covering 3,946 people currently beyond the service radius; nearest working point 11.8 km away.

Chapter 5: Review and Determination

Chapter 5 records the review of the findings in Chapter 4: where a reviewing officer overruled the analysis, what each such decision cost in coverage, and the outcome of the checks applied independently of the analysis.

Reviewing Officer Determinations

Each determination below was applied on its own and the plan re-solved, so that each carries its own cost rather than one aggregate figure. No determination was blocked. A determination excluding a location excludes an area around it, not a single lattice point.

Table 7. Reviewing Officer Determinations

Determination	Site	Reason recorded	Coverage cost
RESCOPE	run scope	The marginal-coverage curve for the first 10 sites hasn't flattened: site 10 still adds 4,207 people, close to site 7's 4,609. Before settling the capital ask at 10, I want the curve out to 14 to see whether it's worth asking for more before this programme closes, per the original budget decision's own uncertainty about the ideal number.	+16,257

Effect of Determinations on Coverage

Overrides reach 16,257 more people than the unconstrained plan. Greedy selection is not optimal, so a constraint can occasionally improve it.

Independent Checks

Table 8. Checks Applied Before Issue

Check	Result	Finding
coverage arithmetic	PASS	union recount matches at 448,414; across the 14 new sites a per-facility sum would report 85,726 against a true union of 84,860, an overcount of 867
geometry	PASS	all 14 sites lie inside the area of interest
budget	PASS	14 sites within the budget of 14
coordinate reference system	PASS	EPSG:32636 (WGS 84 / UTM zone 36N); over 400 sampled pairs the projected distance departs from great-circle by 0.317% at the median and 0.595% at most, or about 6 m on a 1000 m service radius
data currency	FLAG	median source record is 1.9 years old. The age was put to the accountable officer and accepted: conditional, some will have been repaired and some of the 354 will have failed
aggregation sensitivity	PASS	reweighting the demand surface within cells moves the coverage claim from 78.6% to 78.5%, a shift of 0.06%
boundary effect	FLAG	4 of 14 sites lie within one service radius of the district boundary; the nearest is 625 m from it. Population and facilities across the boundary are not represented, so coverage at these sites may be misstated
equity	PASS	53% of newly covered people live in the densest quartile of cells, against 42% district-wide
provenance	PASS	4 sources recorded, all figures traceable
cartographic consistency	PASS	3 figures reviewed against the account; all consistent

2 of the 10 checks returned a flag rather than a pass. A flag does not prevent issue, and the finding is reproduced above in the terms in which it was raised. It is not resolved in this assessment.

Scoring Review

Before issue, the account of this assessment was scored by a reviewer that receives the results file and nothing else: not the solver, not the code path, and not the reasons recorded in Section 5.1. It scores the account given, not the intention behind the plan.

Table 9. Review Scores Against Their Floors

Dimension	Score	Floor	Weight	Outcome
data adequacy	8.5	5.0	0.25	met
method fitness	7.5	5.0	0.20	met
spatial rigour	9.0	6.0	0.25	met
accountability	10.0	6.0	0.20	met
actionability	9.5	4.0	0.10	met

The weighted score is 8.82 against a floor of 6.50. The assessment is issued only when the weighted score and every individual floor are met. Reviewer: deterministic rules over results.json.

Gate Decisions

A gate never silently permits. Each row below is a point at which the run either proceeded and recorded why that was safe, or refused. A refusal removes a source from the analysis and is stated here rather than left to be inferred from a missing figure.

Table 10. Safety Gate Decisions

Gate	Outcome	Decided by	Reason recorded
source authorisation	allowed	harness	Water Point Data Exchange Plus (WPdx+) is a public endpoint; no authorisation required
source authorisation	allowed	harness	WorldPop gridded population is a public endpoint; no authorisation required
guardrail breach	allowed	harness	Overrides reach 16,257 more people than the unconstrained plan. Greedy selection is not optimal, so a constraint can occasionally improve it.

Chapter 6: Sensitivity

Chapter 6 tests how far the recommendation at Chapter 4 depends on assumptions that a reviewing officer may reasonably wish to change. Each scenario re-solves from the same retrieved data with a single assumption altered.

Table 11. Sensitivity of the Recommendation

Scenario	Covered	Share	Against base case
Rehabilitation of existing points only	400,081	70.1%	–32,076 people
Half the budget (7 sites)	418,828	73.4%	–13,329 people
Double the budget (28 sites)	492,169	86.2%	+60,012 people

Rehabilitation Against New Construction

The base case does not select rehabilitation of existing water points, because a location recorded as not serving generally stands beside one that is serving and therefore adds little new coverage. Rehabilitation is nonetheless materially cheaper per unit than new construction, and cost is not modelled in this assessment. Table 8 states what a rehabilitation-only programme of the same size would achieve, so the two can be weighed against costs held by the district.

Appendix A: Data Anomalies and Their Handling

Each item below is a property of the published source data that would have changed a figure in this assessment had it been read differently. Each is recorded with what was observed, what was done, and what a reader must hold in mind as a result. None were resolved silently.

A.1 SEMANTICS · WPdx+

Observed	status_clean is not two-valued in this district. Values present: Non-Functional (891); Functional, needs repair (354); Functional, not in use (6).
Handling	Serving status is read from the status_semantics block in handbooks/wpdx.yaml, which treats Functional and Functional, needs repair as serving and everything else as not serving.
Bearing	An equality test against the string Functional would score every point in this district as not serving and inflate the gap.

A.2 DUPLICATION · WPdx+

Observed	127 of 1,378 returned records were removed: flagged duplicate by WPdx.
Handling	Removed before any coverage computation, per the cleaning rules declared in the handbook.
Bearing	Coverage computed on the raw response would count these records twice or count superseded observations.

A.3 METHOD · MAP friction surface

Observed	Over this area the walking friction surface implies 21 minutes per kilometre at the median and 60 at the 90th percentile, so the chosen 1000 m corresponds to about 21 minutes of walking.
Handling	Coverage is tested as walking time rather than straight-line distance. Reach is computed per candidate from the local friction value rather than by a least-cost accumulation from every candidate, which would be exact but is not tractable at this candidate count.
Bearing	Where terrain changes sharply within one facility's reach, the local-friction approximation applies the conditions at the site in every direction, and is conservative rather than optimistic. The surface is modelled at about one kilometre and does not represent local footpaths.

Appendix B: Retrieval Record

Every retrieval made for this assessment, with the query as executed.

WPdx+

```
clean_country_name = 'Uganda' AND clean_adm2 = 'Mayuge'
```

Endpoint <https://data.waterpointdata.org/resource/eqje-vguj.json>. Retrieved 2026-09-04T10:53:29Z, and read from this machine's local cache for the present run rather than requested again. Returned 1,378 records, retained 1,251. Licence: CC BY 4.0.

GADM

```
UGA administrative level 1
```

Endpoint https://geodata.ucdavis.edu/gadm/gadm4.1/json/gadm41_UGA_1.json.zip. Retrieved 2026-09-03T01:51:52Z, and read from this machine's local cache for the present run rather than requested again. Returned 58 records, retained 58. Licence: GADM licence, free for academic and non-commercial use.

WorldPop

```
UGA ppp 2020, bbox 33.2835 -0.0474 33.7104 0.6061 (WGS 84), aggregated 5x5
```

Endpoint <https://hub.worldpop.org/rest/data/pop/wpgp>. Retrieved 2026-09-03T01:59:46Z, and read from this machine's local cache for the present run rather than requested again. Returned 15,912 records, retained 8,992. Licence: CC BY 4.0.

MAP friction surface

```
Accessibility__202001_Global_Walking_Only_Friction_Surface, bbox 33.2335 -0.0974 33.7604 0.6561 (WGS 84)
```

Endpoint <https://data.malariaatlas.org/geoserver/Accessibility/ows>. Retrieved 2026-09-04T10:53:31Z, and read from this machine's local cache for the present run rather than requested again. Returned 5,670 records, retained 5,670. Licence: CC BY 4.0.

Appendix C: Reproduction Record

This assessment is a function of one results file and the manifest that produced it. Replaying the command below against the recorded queries regenerates it.

```
python -m siting.cli --country Uganda --adm2 Mayuge --iso3 UGA --domain water --mode manual
--out sessions/mayuge-20260904T152614Z-6127d5/runs --format both --reviewer rules --
decisions sessions/mayuge-20260904T152614Z-6127d5/decisions.yaml --overrides overrides/
mayuge.yaml --figure-review sessions/mayuge-20260904T152614Z-6127d5/runs/mayuge-
water-20260904T155905Z/figure_review.json --resume mayuge-water-20260904T155905Z
```

Provenance hash `ac15a4eae9de9c52`. Any change to a source query, a cleaning rule or a retrieval date changes this hash.

Determinations file as supplied

```
# Planner overrides for Mayuge, water domain.
# Every entry needs a reason. The reason is printed in Ex07 alongside what
# the decision cost in coverage.

- verb: RESCOPE
  budget: 14
  actor: Jadey
  reason: >-
    The marginal-coverage curve for the first 10 sites hasn't flattened: site
    10 still adds 4,207 people, close to site 7's 4,609. Before settling the
    capital ask at 10, I want the curve out to 14 to see whether it's worth
    asking for more before this programme closes, per the original budget
    decision's own uncertainty about the ideal number.
```

Table 12. Sources, Licences and Access Dates

Source	Licence	Accessed
Water Point Data Exchange (WPdx+), global water point registry, https://www.waterpointdata.org/	CC BY 4.0	2026-09-04
GADM database of Global Administrative Areas, version 4.1, https://gadm.org	GADM licence, free for academic and non-commercial use	2026-09-03
WorldPop, School of Geography and Environmental Science, University of Southampton, https://www.worldpop.org	CC BY 4.0	2026-09-03
Weiss, D.J. et al. Global maps of travel time to healthcare facilities. <i>Nature Medicine</i> 26, 1835-1838 (2020).	CC BY 4.0	2026-09-04